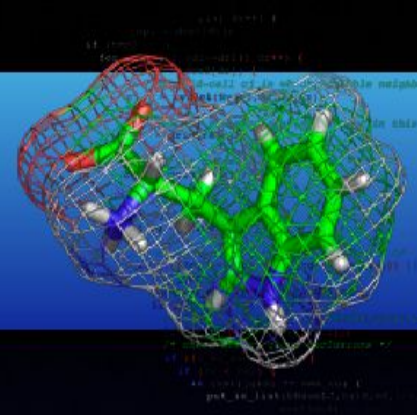


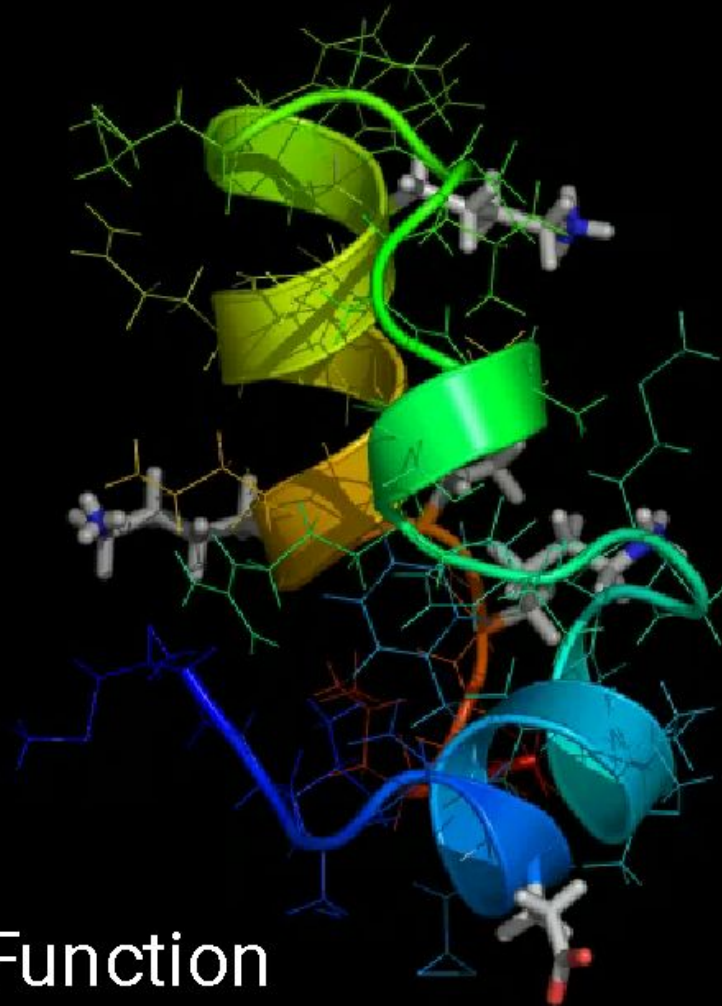
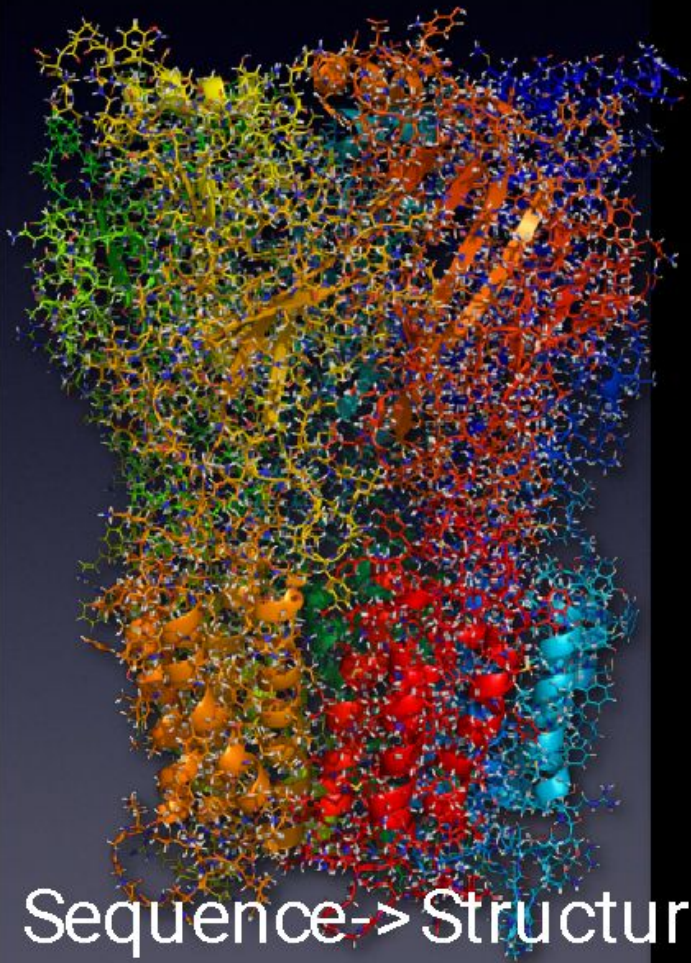
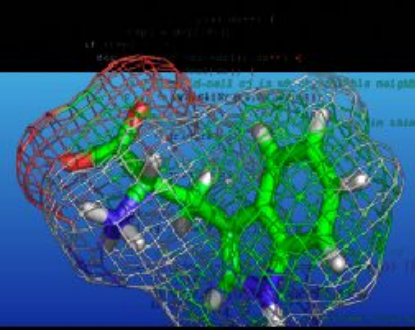
Target: Protein (Continued)

Outline today



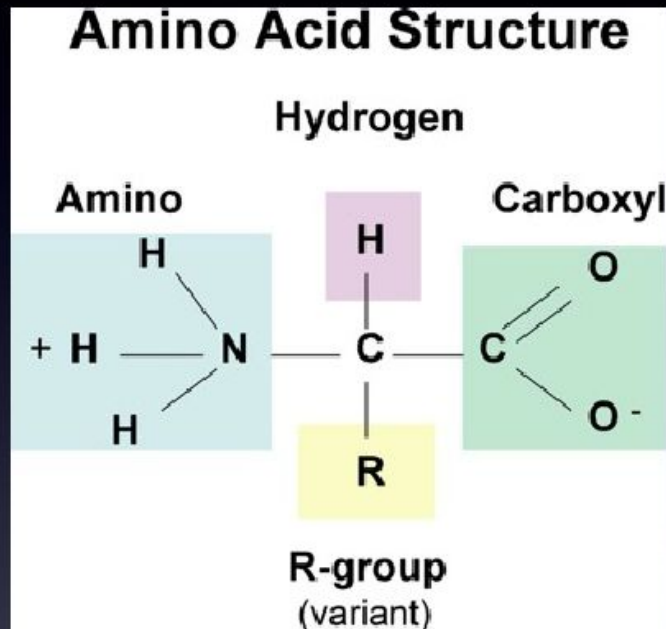
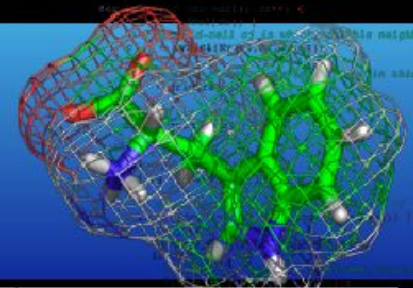
- Basic concepts - possibly repetition for some
- Amino acids, proteins
- Basic physical properties of aa & proteins
- Structure of proteins, protein folding
- Elementary interactions in proteins

Protein structure

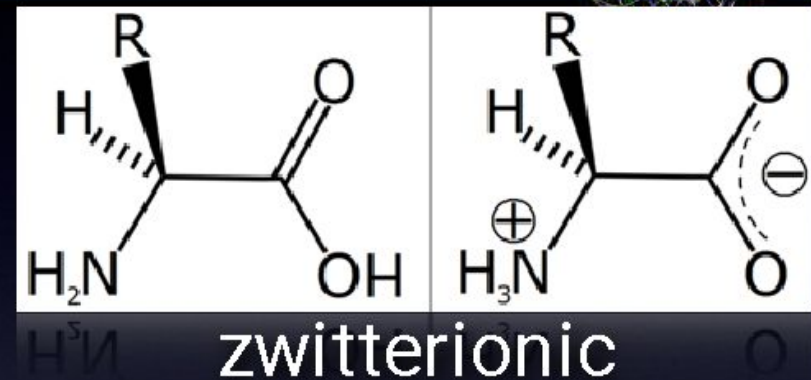


Sequence-→ Structure-→ Function

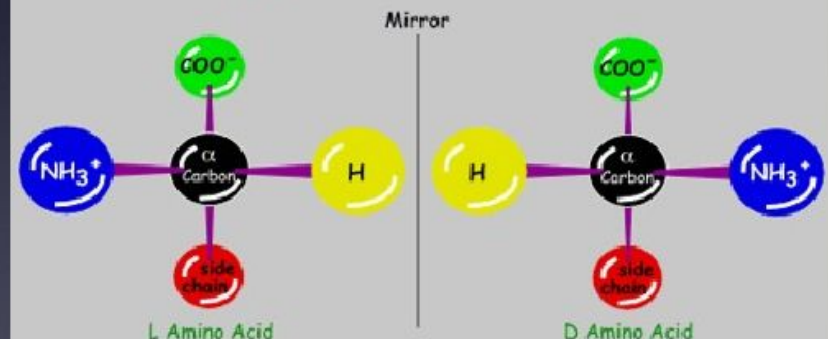
Amino acids



different properties



Common Amino Acids are Stereoisomers,
Meaning they have a Chiral α Carbon center.



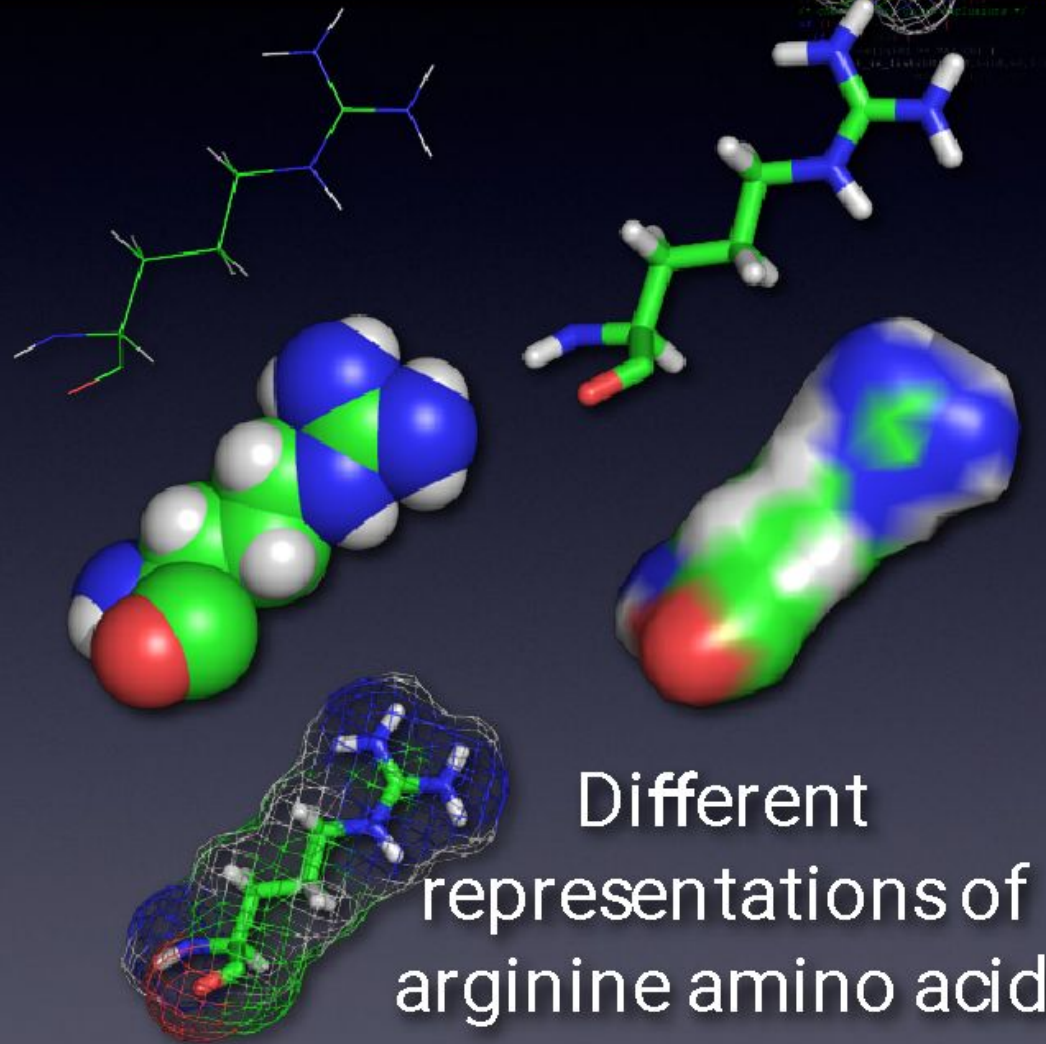
Amino Acids can exist in either the D or L configuration.
However, All Chiral Amino Acids in Proteins have the L configuration

19 out of 20 are chiral: natural state is L!

Amino acids

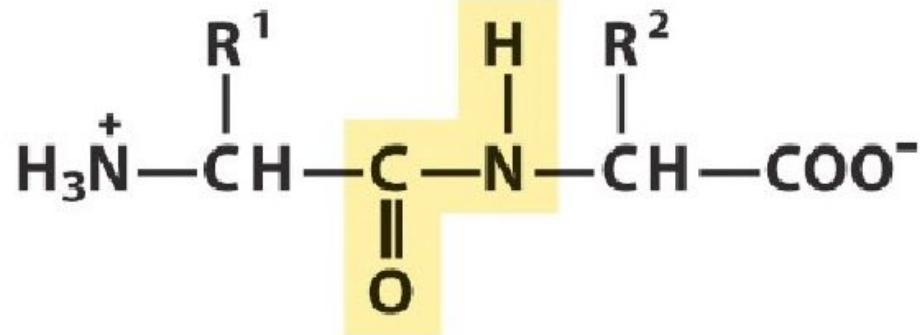
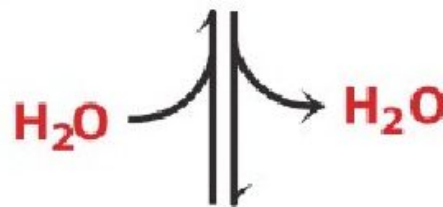
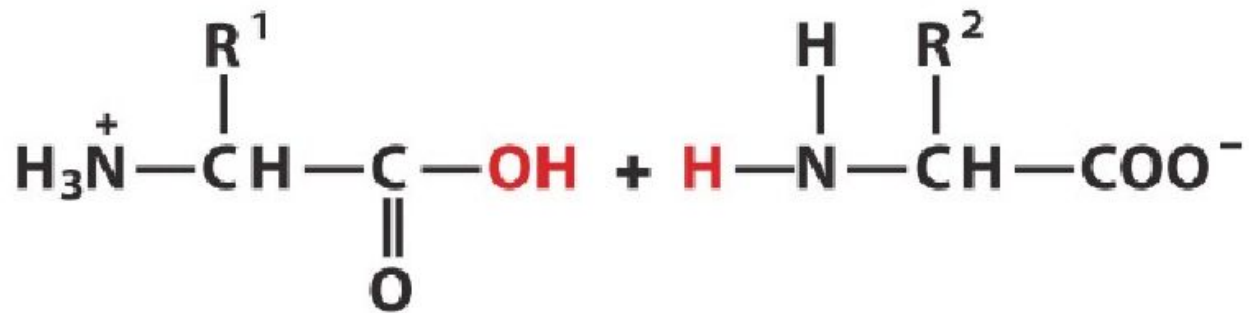
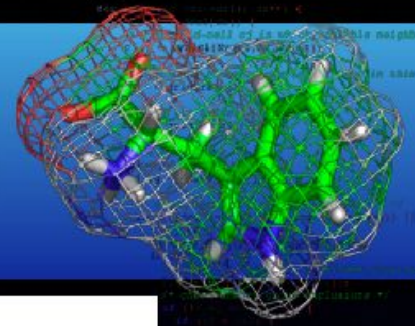
NONPOLAR, HYDROPHOBIC		POLAR, UNCHARGED	
	R GROUPS		
Alanine Ala A MW = 89	$\begin{array}{c} \text{H}_3\text{N}^+ \\ \\ \text{CH} - \text{CH}_3 \\ \\ \text{COO}^- \end{array}$	$\begin{array}{c} \text{H} - \text{CH} - \text{COO}^- \\ \\ \text{NH}_3^+ \end{array}$	Glycine Gly G MW = 75
Valine Val V MW = 117	$\begin{array}{c} \text{H}_3\text{N}^+ \\ \\ \text{CH} - \text{CH}(\text{CH}_3)_2 \\ \\ \text{COO}^- \end{array}$	$\begin{array}{c} \text{HO-CH}_2 - \text{CH} - \text{COO}^- \\ \\ \text{NH}_3^+ \end{array}$	Serine Ser S MW = 105
Leucine Leu L MW = 131	$\begin{array}{c} \text{H}_3\text{N}^+ \\ \\ \text{CH} - \text{CH}_2 - \text{CH}(\text{CH}_3)_2 \\ \\ \text{COO}^- \end{array}$	$\begin{array}{c} \text{OH-CH}(\text{CH}_3) - \text{CH} - \text{COO}^- \\ \\ \text{NH}_3^+ \end{array}$	Threonine Thr T MW = 119
Isoleucine Ile I MW = 131	$\begin{array}{c} \text{H}_3\text{N}^+ \\ \\ \text{CH} - \text{CH}(\text{CH}_3) - \text{CH}_2 - \text{CH}_3 \\ \\ \text{COO}^- \end{array}$	$\begin{array}{c} \text{HS-CH}_2 - \text{CH} - \text{COO}^- \\ \\ \text{NH}_3^+ \end{array}$	Cysteine Cys C MW = 121
Phenylalanine Phe F MW = 131	$\begin{array}{c} \text{H}_3\text{N}^+ \\ \\ \text{CH} - \text{CH}_2 - \text{C}_6\text{H}_5 \\ \\ \text{COO}^- \end{array}$	$\begin{array}{c} \text{HO-C}_6\text{H}_4 - \text{CH}_2 - \text{CH} - \text{COO}^- \\ \\ \text{NH}_3^+ \end{array}$	Tyrosine Tyr Y MW = 181
Tryptophan Trp W MW = 204	$\begin{array}{c} \text{H}_3\text{N}^+ \\ \\ \text{CH} - \text{CH}_2 - \text{C}_8\text{H}_6\text{N} \\ \\ \text{COO}^- \end{array}$	$\begin{array}{c} \text{NH}_2 - \text{C}(=\text{O}) - \text{CH}_2 - \text{CH} - \text{COO}^- \\ \\ \text{NH}_3^+ \end{array}$	Asparagine Asn N MW = 132
Methionine Met M MW = 149	$\begin{array}{c} \text{H}_3\text{N}^+ \\ \\ \text{CH} - \text{CH}_2 - \text{CH}_2 - \text{S-CH}_3 \\ \\ \text{COO}^- \end{array}$	$\begin{array}{c} \text{NH}_2 - \text{C}(=\text{O}) - \text{CH}_2 - \text{CH}_2 - \text{CH} - \text{COO}^- \\ \\ \text{NH}_3^+ \end{array}$	Glutamine Gln Q MW = 146
Proline Pro P MW = 115	$\begin{array}{c} \text{H}_3\text{N}^+ \\ \\ \text{CH} - \text{CH}_2 - \text{CH}_2 \\ \quad \\ \text{HN} \quad \text{CH}_2 \end{array}$	POLAR BASIC	
Aspartic acid Asp D MW = 133	$\begin{array}{c} \text{H}_3\text{N}^+ \\ \\ \text{CH} - \text{CH}_2 - \text{C}(=\text{O})\text{O}^- \\ \\ \text{COO}^- \end{array}$	$\begin{array}{c} \text{NH}_2 - \text{C}(=\text{O}) - \text{NH} - (\text{CH}_2)_3 - \text{CH} - \text{COO}^- \\ \\ \text{NH}_3^+ \end{array}$	Lysine Lys K MW = 146
Glutamic acid Glu E MW = 147	$\begin{array}{c} \text{H}_3\text{N}^+ \\ \\ \text{CH} - \text{CH}_2 - \text{CH}_2 - \text{C}(=\text{O})\text{O}^- \\ \\ \text{COO}^- \end{array}$	$\begin{array}{c} \text{H}_2\text{N} - \text{C}(=\text{O}) - \text{CH} = \text{CH} - \text{CH} - \text{COO}^- \\ \\ \text{NH}_3^+ \end{array}$	Histidine His H MW = 155

atomic vs molecular
levels

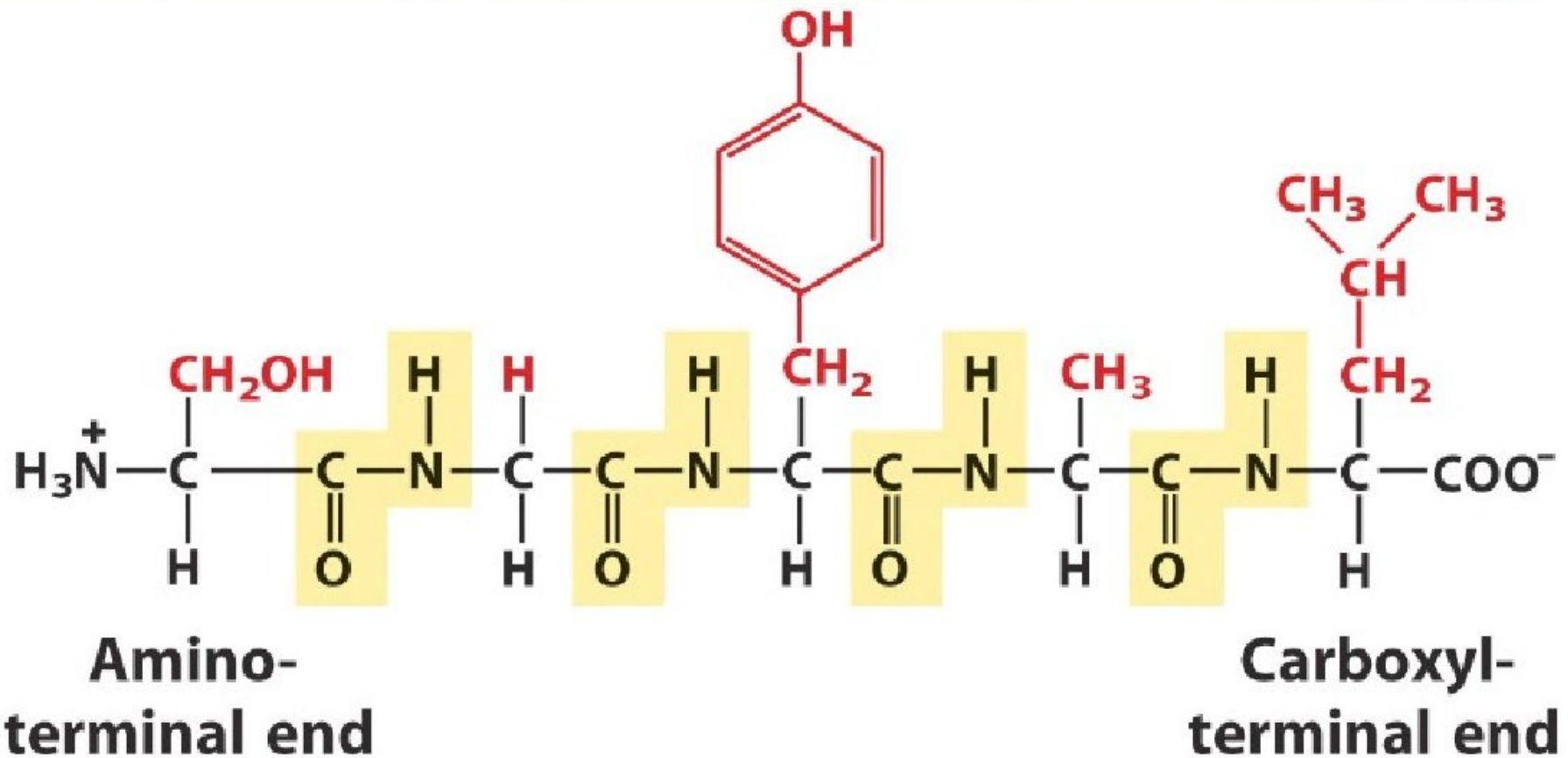
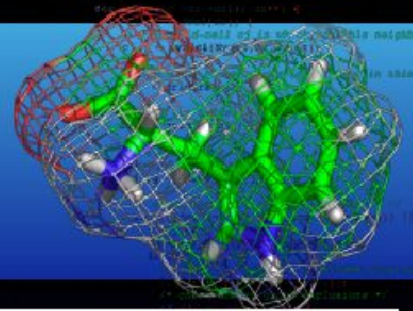


POLAR AMINO ACIDS				NONPOLAR AMINO ACIDS			
AMINO ACID		SIDE CHAIN		AMINO ACID		SIDE CHAIN	
Aspartic acid	Asp	D	negative	Alanine	Ala	A	nonpolar
Glutamic acid	Glu	E	negative	Glycine	Gly	G	nonpolar
Arginine	Arg	R	positive	Valine	Val	V	nonpolar
Lysine	Lys	K	positive	Leucine	Leu	L	nonpolar
Histidine	His	H	positive	Isoleucine	Ile	I	nonpolar
Asparagine	Asn	N	uncharged polar	Proline	Pro	P	nonpolar
Glutamine	Gln	Q	uncharged polar	Phenylalanine	Phe	F	nonpolar
Serine	Ser	S	uncharged polar	Methionine	Met	M	nonpolar
Threonine	Thr	T	uncharged polar	Tryptophan	Trp	W	nonpolar
Tyrosine	Tyr	Y	uncharged polar	Cysteine	Cys	C	nonpolar

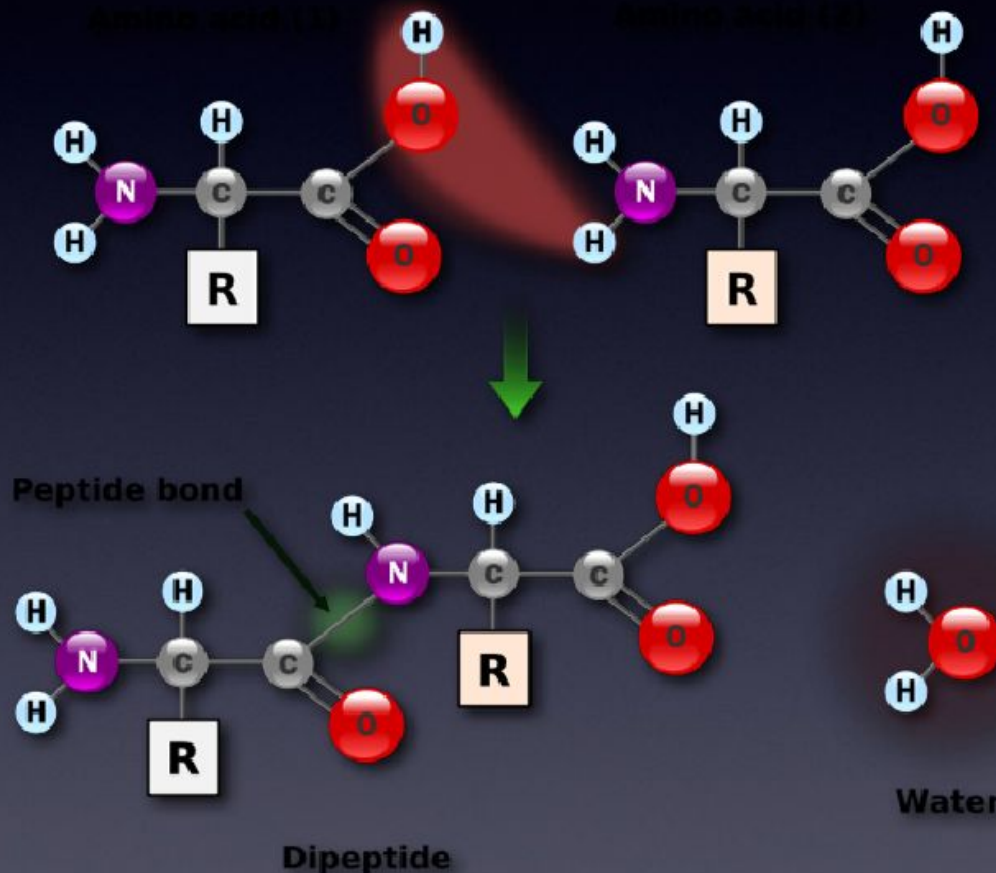
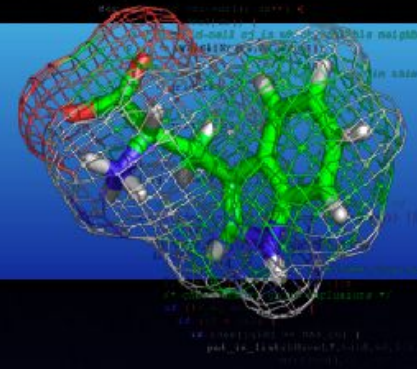
Polymerization



Polypeptides



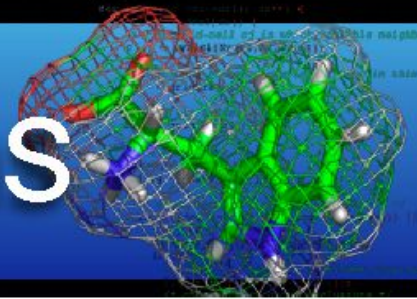
Peptide bonds



Peptide bond
rigid & planar
(sp²-hybridization,
electron
delocalized)

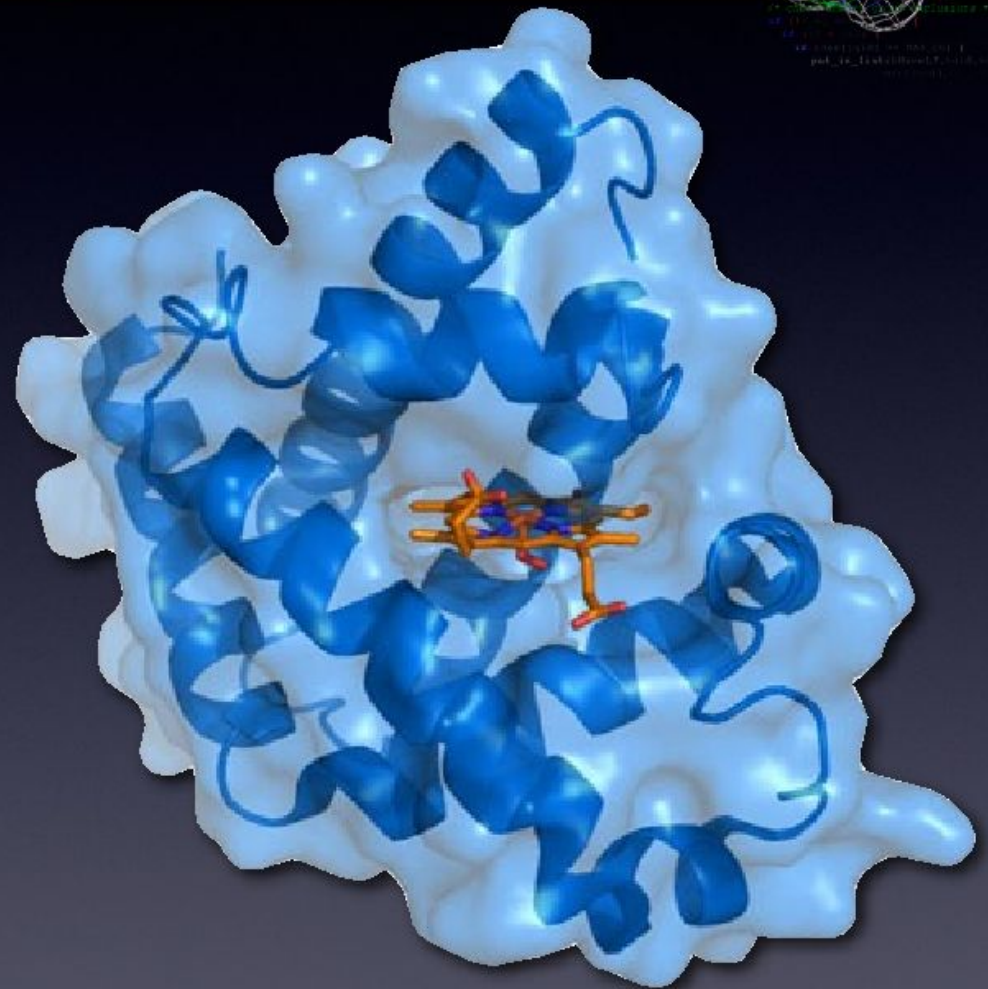
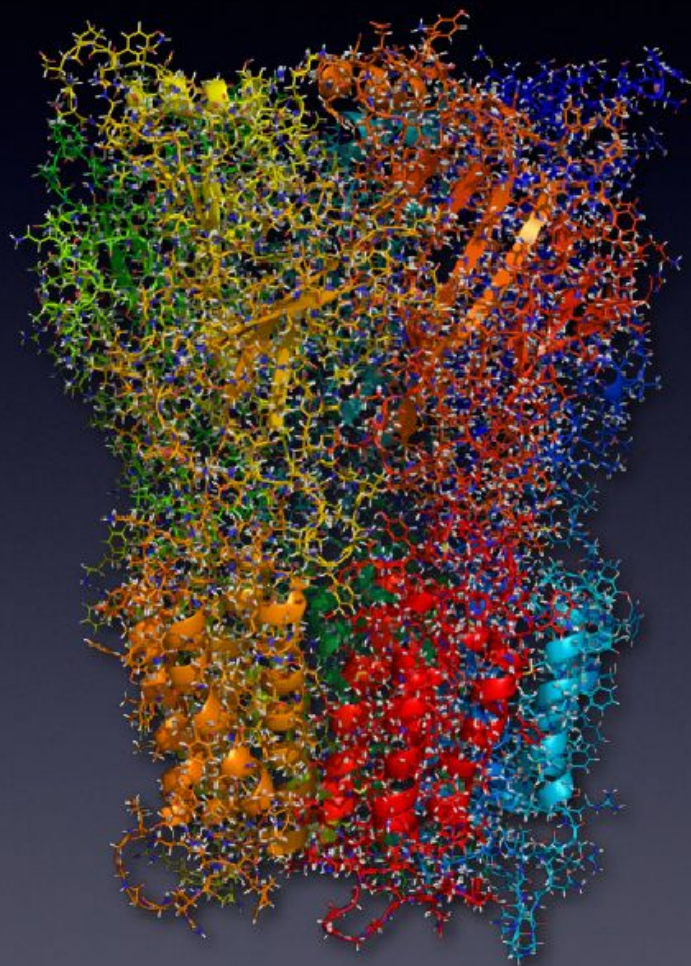
Peptide bond
polarity

Example Proteins

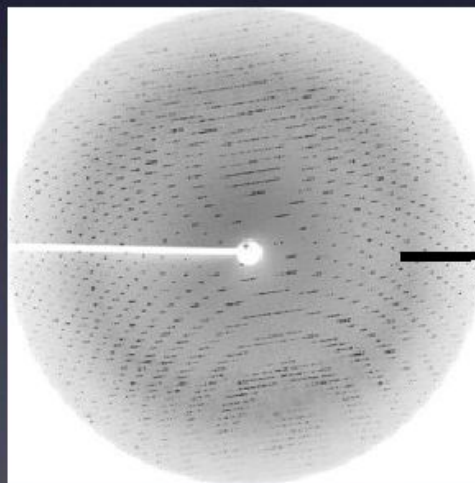
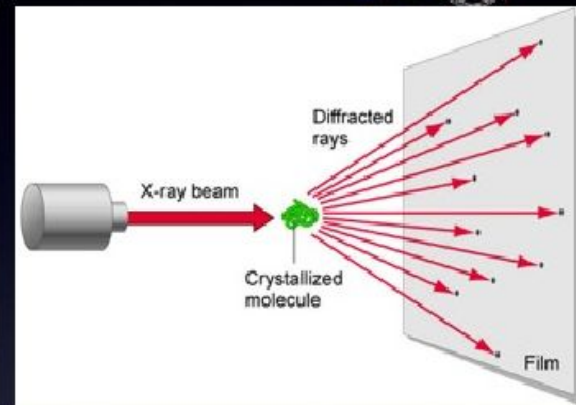
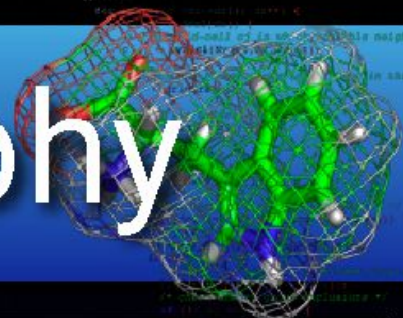


weight ~110 per amino acid	Molecular weight	Number of residues	Number of polypeptide chains
Cytochrome c (human)	13,000	104	1
Ribonuclease A (bovine pancreas)	13,700	124	1
Lysozyme (chicken egg white)	13,930	129	1
Myoglobin (equine heart)	16,890	153	1
Chymotrypsin (bovine pancreas)	21,600	241	3
Chymotrypsinogen (bovine)	22,000	245	1
Hemoglobin (human)	64,500	574	4
Serum albumin (human)	68,500	609	1
Hexokinase (yeast)	102,000	972	2
RNA polymerase (<i>E. coli</i>)	450,000	4,158	5
Apolipoprotein B (human)	513,000	4,536	1
Glutamine synthetase (<i>E. coli</i>)	619,000	5,628	12
Titin (human)	2,993,000	26,926	1

An assembled protein

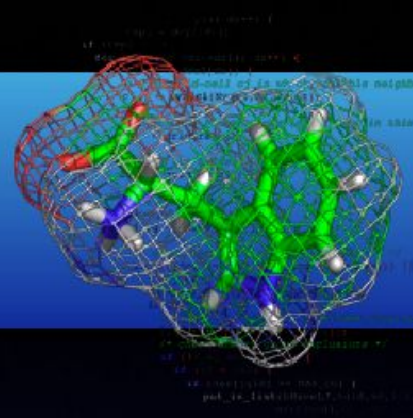


X-ray crystallography



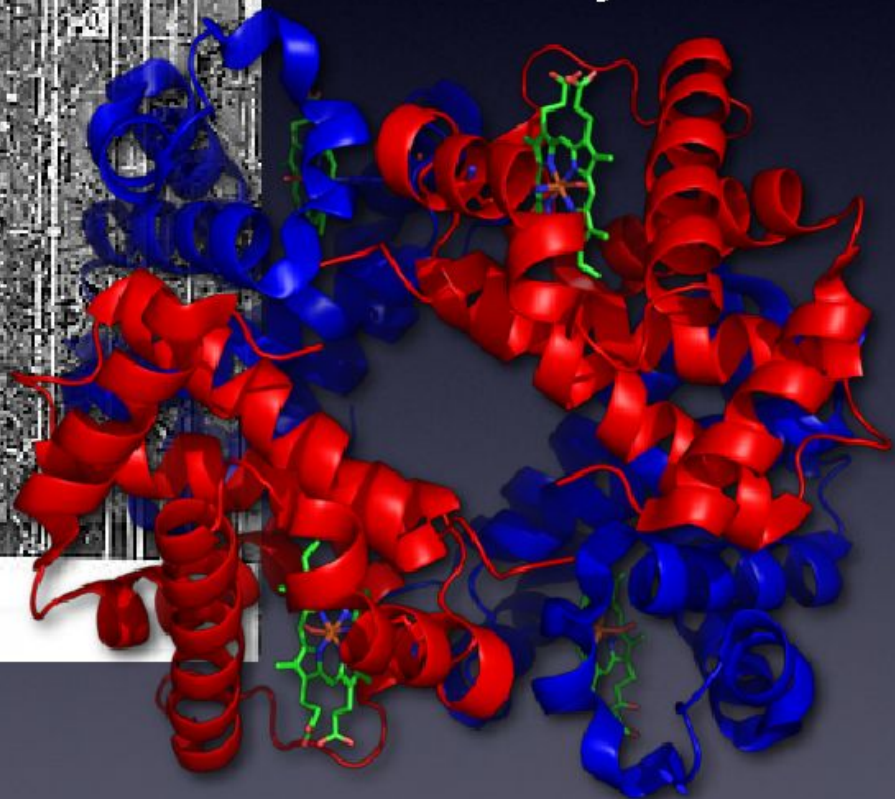
Protein
structure

Protein structure

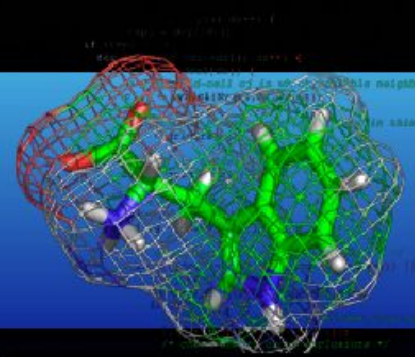


X-ray structure
Took 22 years...

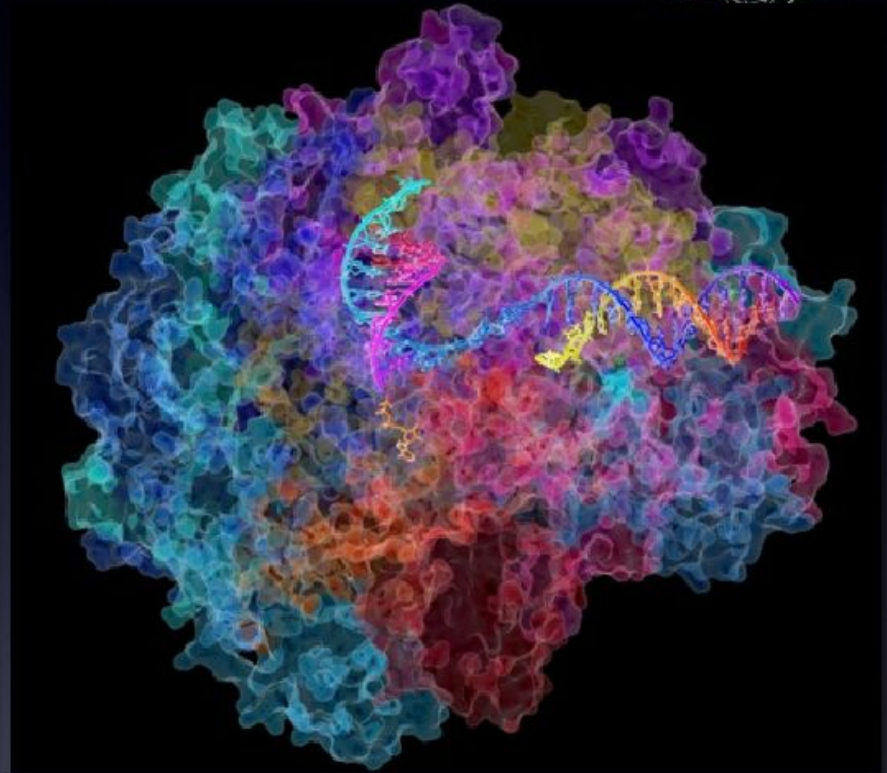
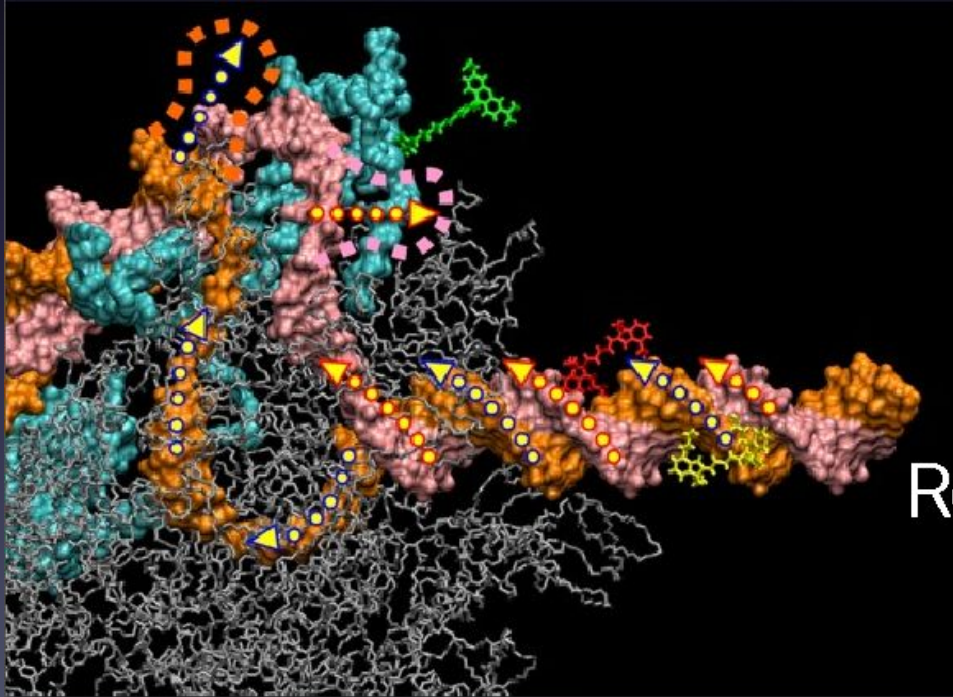
Max Perutz & Hemoglobin



Protein structure

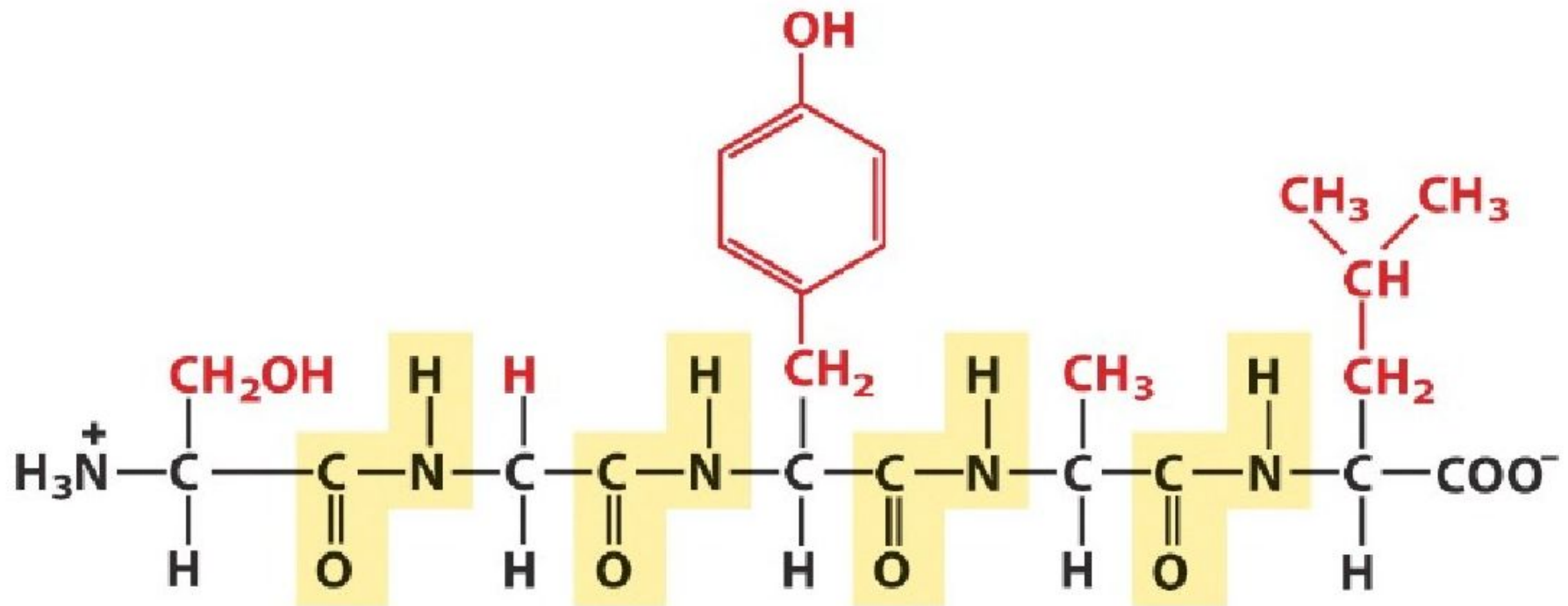
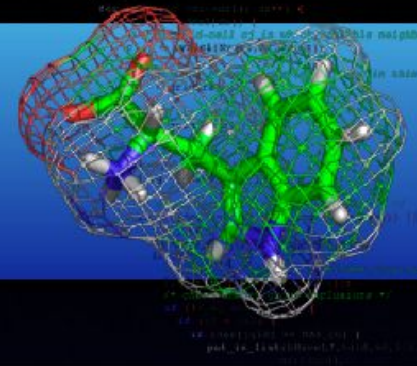


RNA Polymerase



Roger Kornberg 2006

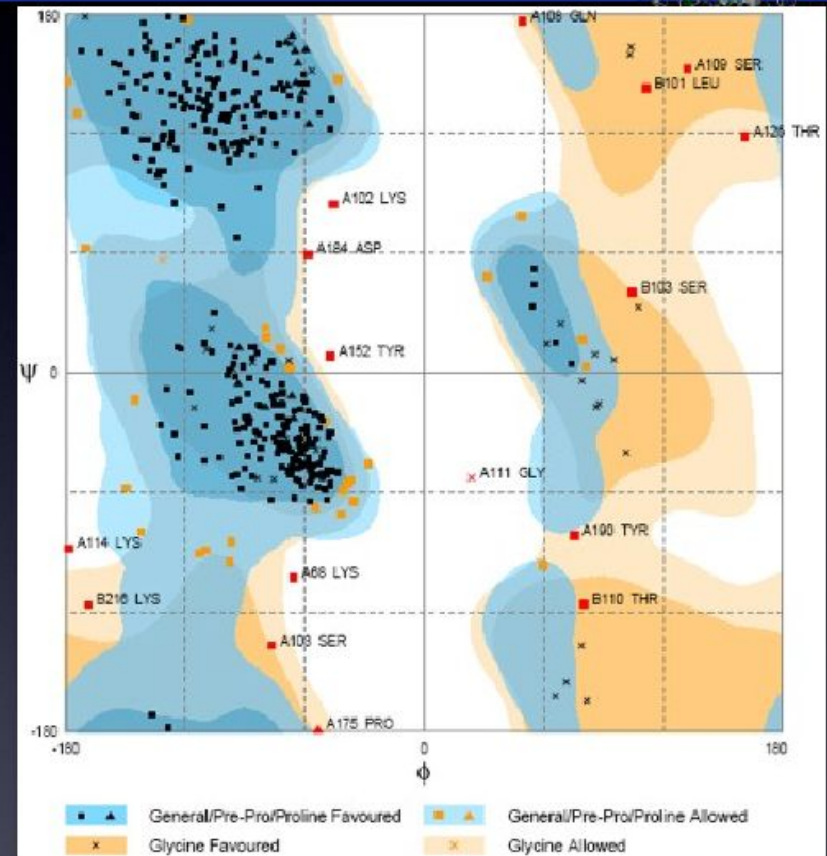
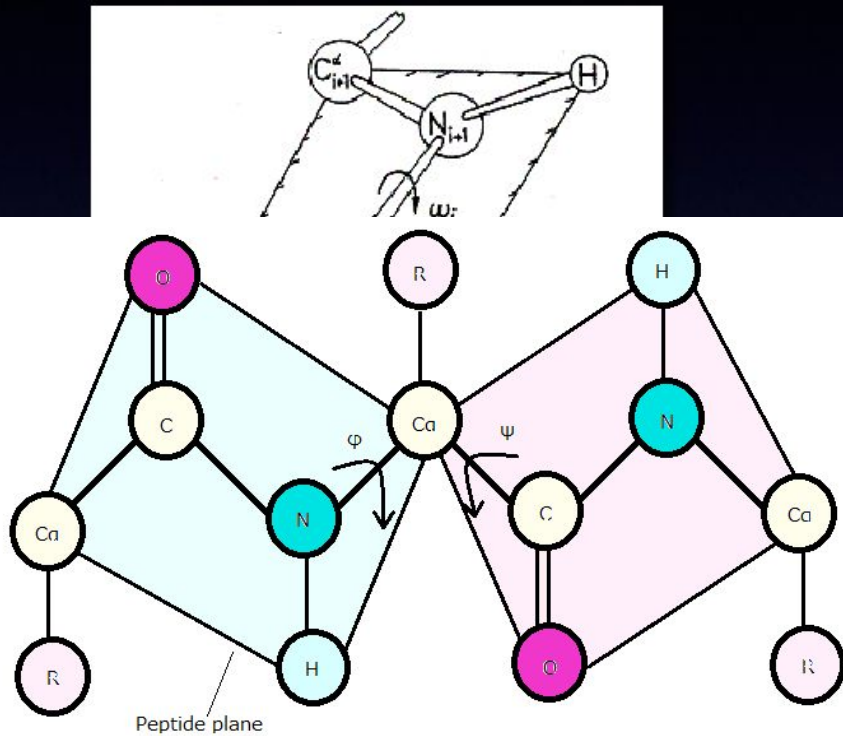
Why this diversity?



Amino-
terminal end

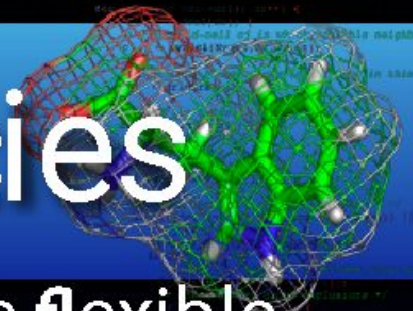
Carboxyl-
terminal end

Ramachandran diagrams



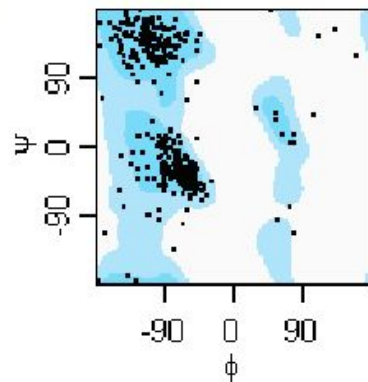
Natural degrees of freedom for amino acids
(due to steric clashes)

Ramachandran species

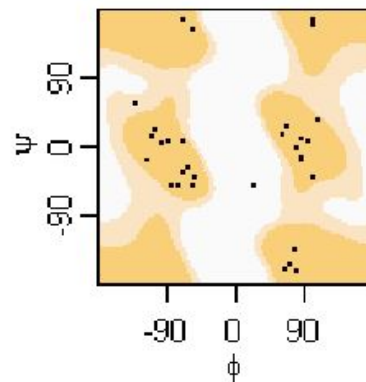


Glycine more flexible
Proline less flexible;
even messes up
previous residue!

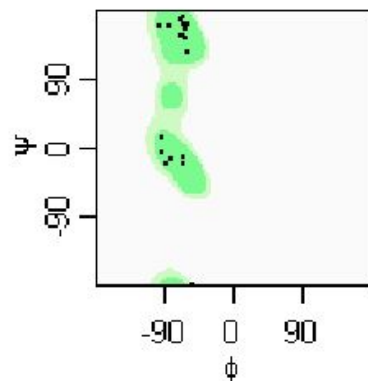
General



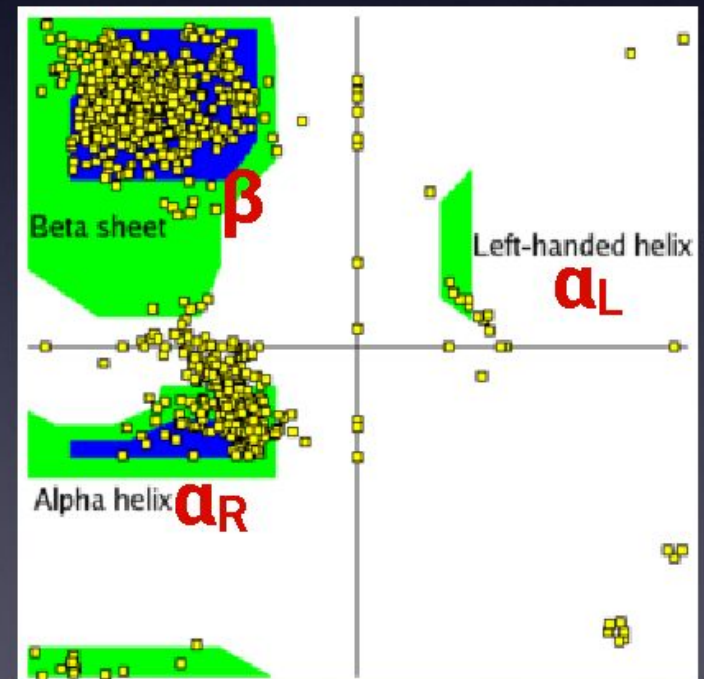
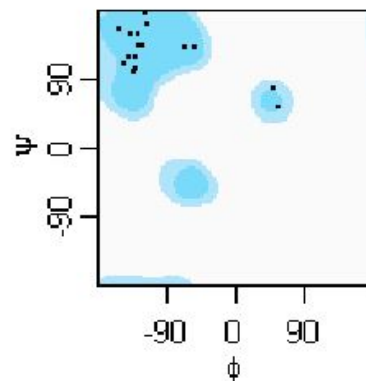
Glycine



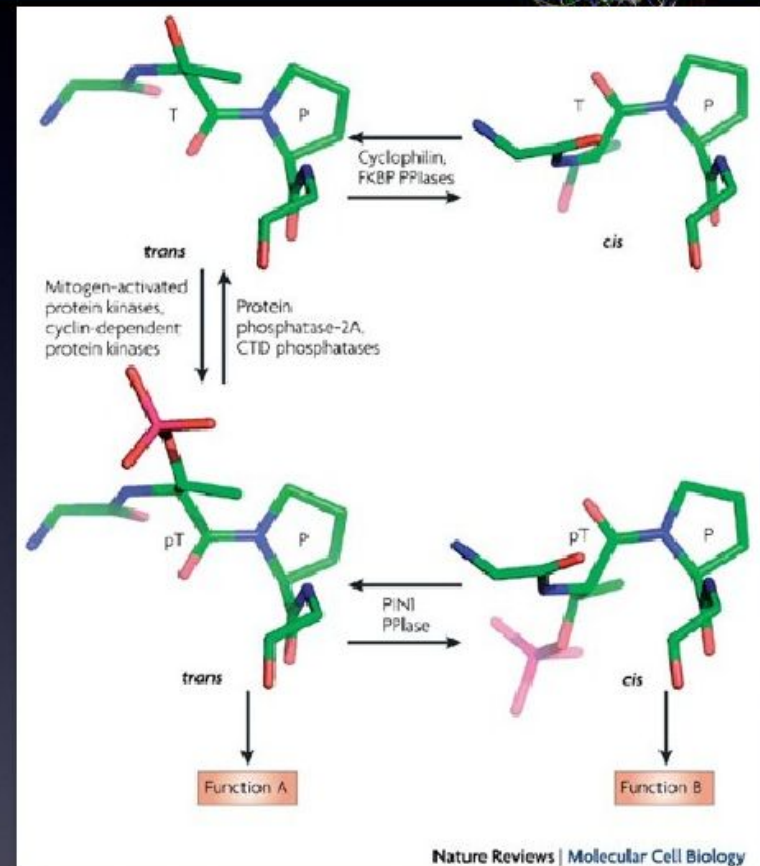
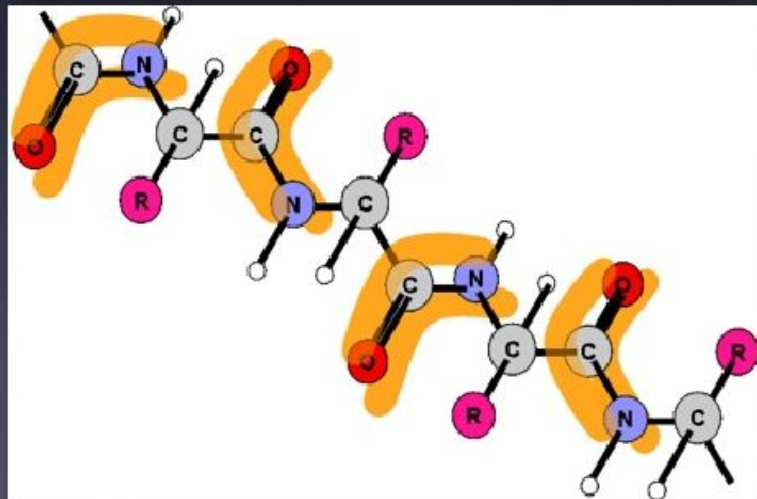
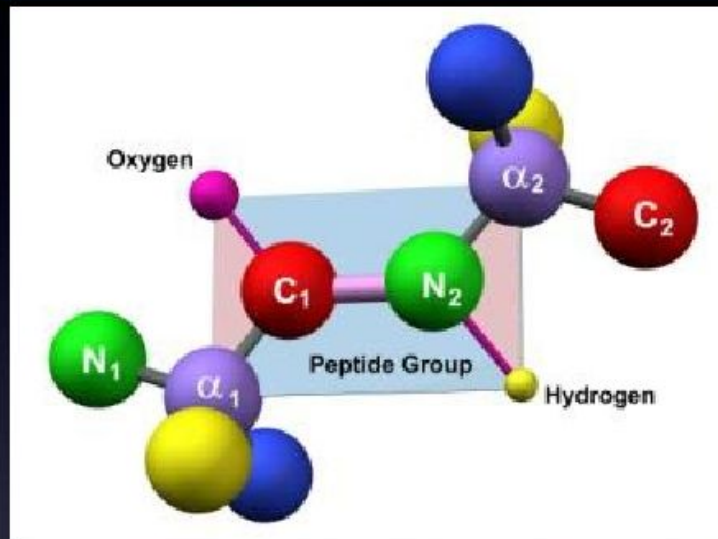
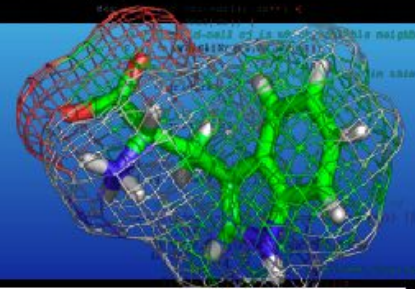
Proline



Pre-Pro

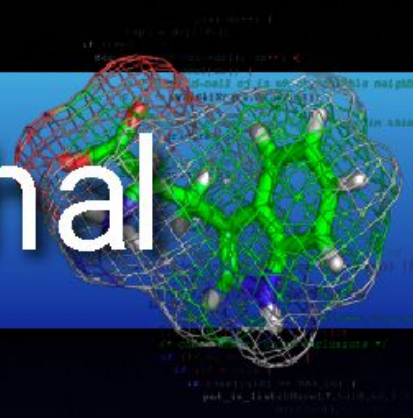


Cis/trans isomers



In Proline, both Cis & Trans occur, and it is important for function!

Anfinsen & Levinthal



- Anfinsen: Proteins adopt the structure corresponding to the global minimum in free energy
- Levinthal: But how can a protein find that? Even with just 2 states per amino acid and 100 residues there are 2^{100} states! (*Levinthal's Paradox*)

100 amino acids

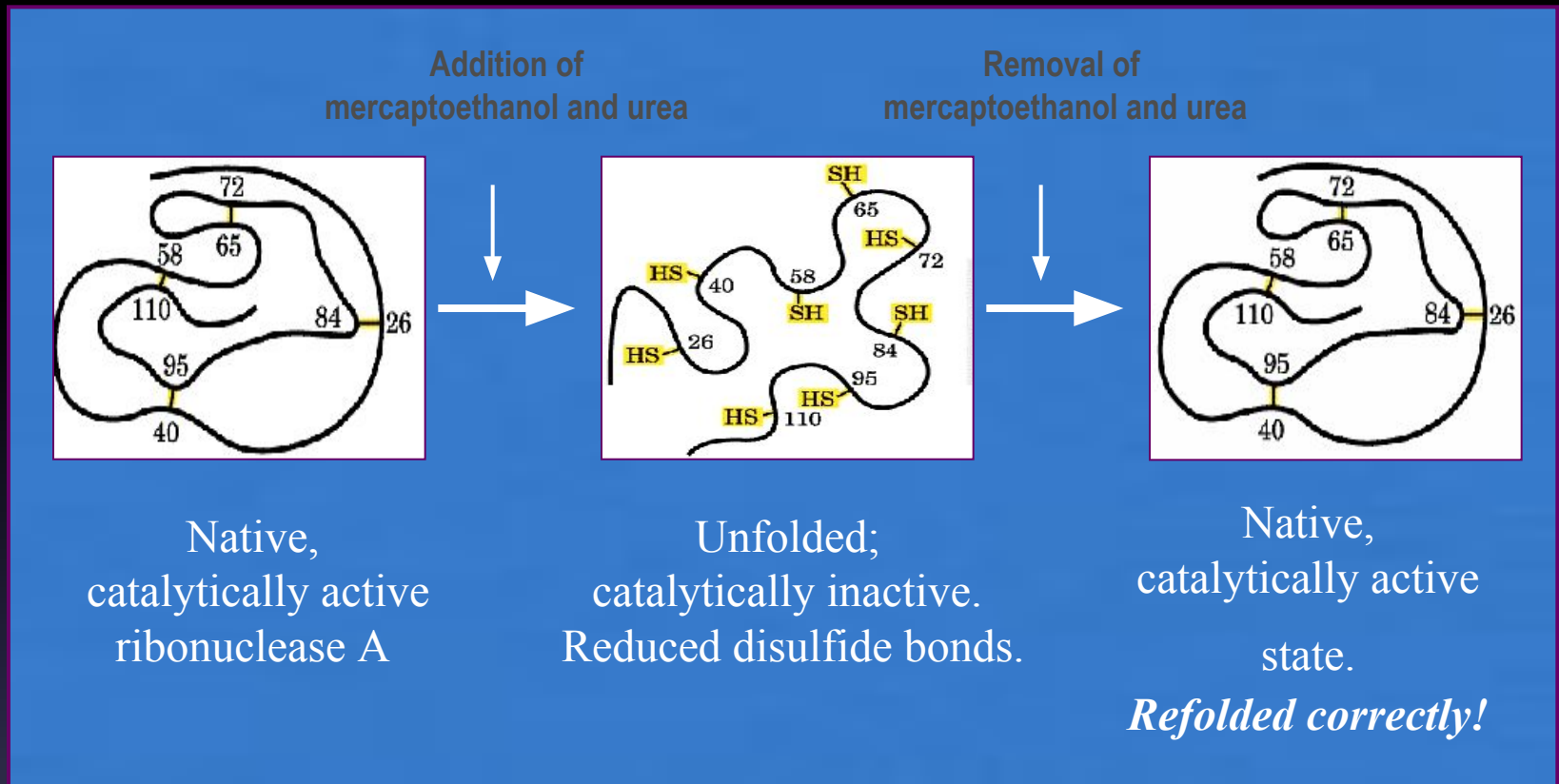
2 states (psi and phi)

2×360 states

1 state = 1 sec

2×360 sec $\rightarrow$ 1 amino acid

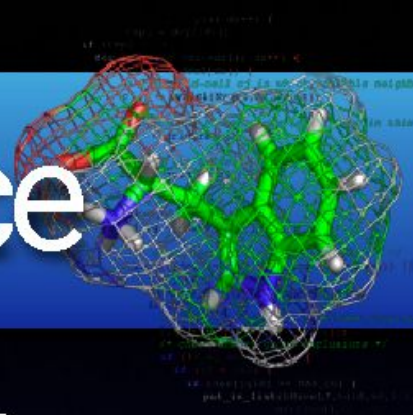
Anfinsen's Experiment



**1/105 random
chance**

***Folding is encoded in the amino acid sequence.
Native state is the minimum energy state.***

Conformational space



- How many conformations are there?
- Sample Φ, Ψ torsion in 10 degree units
 - 36 'states' for each torsion
 - 36^2 states per residue
- For a 100-residue chain we get:
 - $(36^2)^{100} = 36^{200} \approx 10^{308}$ states for the chain
- Only one is the native structure